Total No. of Questions : 10] [Total No. of Printed Pages : 4

Paper Code : 21316

F-416

B.C.A. (Fourth Semester)

Examination, 2025

(New Course)

Paper -BCA-(N)-401

Operating System

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any five questions. All questions carry equal marks. Symbol used have their usual meanings.

1. (a) Discriminate between batch operating system and multiprocessor operating system with neat and clean diagram.

   (b) What do you mean by non-preemptive algorithm? Explain it through an example.

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2. (a) Define deadlock. Also Describe deadlock detection methods.

   (b) Write short notes on the following terms:

   (i) Virtual memory

   (ii) FCFS

3. (a) Compare long term, short term and midterm scheduler.

   (b) List the various components of Operating System.

4. (a) Consider the following reference string:

   1, 2, 3, 4, 2, 1, 6, 5.1.2.3,7.6, 3, 2.1.2, 3.6

   How many page faults will occur for:

   (i) LRU replacement

   (ii) Optimal page replacement

   (b) Write the name of disk schedule algorithm. Write the method and explain the working of any two algorithm.

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5. (a) What is Fragmentation Problem? Describe the external and internal fragmentation.

(b) What the basic functions does an operating system perform as a resources manager?

6. (a) What is Process Control Block? Design basic framework of process control block.

(b) Explain in details about the disk storage and disk scheduling.

7. (a) Explain in detail about the mutual exclusion and critical section problem.

(b) For the following processes, draw Gantt Chart to illustrate the execution using,

(i) SJF

(ii) Round Robin

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Also, calculate average waiting time and average turnaround time. (Assumption: A larger priority number has higher priority.

Process	Arrival time	Burst time	Priority
A	0	5	4
B	2	4	2
C	2	2	6
D	4	4	3

8 (a) Explain in detail about the inter process communication and models.

(b) Differentiate between Best fit and Worse fit.

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Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21316

F-416

B.C.A. (Fourth Semester) Examination, 2024

(New Course)

Paper No. : BCA-N-401

OPERATING SYSTEM

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) Define.
   (i) Multiprogramming
   (ii) Multitasking
   (iii) Multiprocessing
   (iv) Time sharing
   (v) Virtual memory

   (b) What are the major activities of an operating system with regard to process management?

2. (a) Define demand paging in memory management with example.

   (b) Define C-scan scheduling in detail.

3. (a) Distinguish between User file directory (UFD) and Master file directory (MFD)

   (b) Explain Virtual memory. How it is implemented?

4. (a) Describe various types of allocation methods with examples.

   (b) What is deadlock? How do you deal with potential deadlock?

5. (a) Describe the various layers of file system.

   (b) Explain various disk scheduling algorithms with illustrations.

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6. (a) Differentiate between multiprogramming with fixed partition and multiprogramming with variable partition.

(b) Explain the deadlock detection algorithm for single and multiple instance type

7. (a) What are the advantages of indexed allocation? How can the index blocks be implemented in the indexed allocation scheme?

(b) Explain various page replacement strategies.

8. (a) Explain the concept of deadlock recovery

(b) Explain the following allocation algorithms

(i) First-Fit

(ii) Best-Fit

(iii) Worst-Fit

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Total No. of Questions : 8] [Total No. of Printed Pages : 4

Paper Code : 21316

F-416

B.C.A. (Fourth Semester) Examination, 2022

(New Course)

Paper No. BCA-N-401

Operating System

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) Explain the evaluation of operating system? Differentiate between Multiprogramming, and Time sharing system? 7

   (b) Discuss the following (i) Types of services provided by OS (ii) User's view and System's view of OS. 7

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2. (a) Perform CPU Scheduling and calculate average waiting time using (i) Non Preemptive SJF (ii) ROUND ROBIN with time quantum 20. 7

Process	Burst Time
P_1	53
P_2	17
P_3	68
P_4	24

   (b) Discuss the following (i) Process Control Block (ii) Types of Schedulers (iii) Context Switch (iv) Process Creation and Termination (v) Hardware and Software Interrupt. 7

3. (a) What do you mean by Semaphore? Discuss implementation of semaphore using Binary Semaphore? 7

   (b) Discuss "Bounded Buffer" Problem and present its solution using Semaphore? 7

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4. (a) What is virtual memory? Explain demand paging. 7

(b) Explain multiprogramming with fixed partition. 7

5. (a) Explain the concept of paging? Discuss the use of TLB for paging scheme? 7

(b) Explain the concept of Segmentation? Discuss the concept of Segments Sharing? https://www.mjpruonline.com 7

6. (a) Discuss Critical-Section Problem? Define various algorithms that provide the solution for the requirements of critical section problem? 7

(b) What do you mean by Page Fault? How many Page Faults would occur for following reference string, for four page frames using FIFO, Optimal algorithm?

7 0 1 2 0 3 0 4 2 3 0 3 2 1 7

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7. (a) Define Disk Scheduling algorithms with an example of each? 7

(b) What do you mean Deadlock Characterization? Discuss various methods for handling Deadlocks? 7

8. Write short note on any **four** of the following: (3.5 each)

(a) Time sharing OS

(b) OS Component

(c) System devices

(d) Goals of protection

(e) File operation

(f) Thrashing

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Paper Code : 21316

F–416

B. C. A. (Fourth Semester) EXAMINATION, 2019

(New Course)

Paper No. BCA–N–401

OPERATING SYSTEM

Time : Three Hours] [*Maximum Marks : 70*

Note : Attempt any *five* questions. All questions carry equal marks.

1. (a) Explain operating system architecture.

   (b) Discuss the similarities and differences between paging and segmentation.

   (c) Compare and contrast multitasking and multiprogramming.

2. (a) Describe the differences among short-term, medium-term and long-term scheduler.

   (b) Explain the term process. What are the various states that a process can undergo ? Also explain PCB with a neat diagram.

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3. (a) With a neat diagram explain Resource Allocation Graph.

   (b) What is deadlock ? Explain the necessary conditions for its occurrence.

4. (a) Let a disk drive has 5000 cylinders from 0 to 4999. Currently drive is at 143rd cylinder, and the previous request was at cylinder 125. Queue of pending request FIFO order 186, 1470, 913, 1774, 948, 1509, 1022, 130. What is total distance the disk arm moves to satisfy all the pending requests for each of the following disk scheduling algorithms from current position ?

   (i) FCFS

   (ii) SCAN

   (iii) LOOK

   (b) What are semaphores ? Explain binary and counting semaphores with an example.

5. (a) Explain with the help of supporting diagram Transition Lead aside Buffers (TLB) improves the performance of demand paging.

   (b) Consider the following reference string 7, 0, 1, 2, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 01 for a memory with three (03) frames. How many page faults access for LRU and FIFO page replacement algorithms ? Which is efficient among both ?

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6. (a) For the following snapshot, find the safe sequence using Banker's algorithm :

	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P_0	0	0	2	0	0	4	1	0	2
P_1	1	0	0	2	0	1			
P_2	1	3	5	1	3	7			
P_3	6	3	2	8	4	2			
P_4	1	4	3	1	5	7			

(i) Is the system in safe state ?

(ii) If a request from process P_2 arrives for (002), can the request be granted immediately ?

(b) Given the memory partitions of 100 k, 500 k, 200 k, 300 k and 600 k apply first fit and last fit algorithm to place 212 k, 417 k, 112 k, 426 k.

7. (a) What is a file ? Explain the different file allocation methods.

(b) What are directories ? List the different directory structures with examples. Mention their advantages and disadvantages.

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